



Shreesha's Python Code Documentation



Day 1 - Python Basics: Printing, Variables, Input/Output, Conditionals, Math Operators



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1. Printing a Multiline Statement

python

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Purpose: Printing a beautiful multi-line welcome message in Kannada.

```
print("""Devare,Nimage Hrutpurvaka Namaskaragalu!  
Indininda Nanu Python Programing Language Kalibeku anta Dhrudavaada Manassannu Madiddene.  
Dayavittu Nimma Ashirvaada&Krupe nanna meleirali,  
yendu Hrudpurvaka Prarthane!!!  
inthe nimma preethiya shishya,  
  
~ Shreesha Krishna Rao""")
```

- **Explanation:**

- Using triple quotes (`"""`) to print multiple lines together.
- No need for `\n` explicitly when using triple quotes.

2. Creating Variables & Checking Types

```
python Copy Edit  
  
# 🚀 Purpose: Creating variables of different types and checking their types.  
  
age = 25          # Integer value  
price = 19.95     # Float value  
name = "Shreesha" # String value  
is_coding = True  # Boolean value  
  
print(age, type(age))          # Prints value and its type  
print(price, type(price))  
print(name, type(name))  
print(is_coding, type(is_coding))
```

- Explanation:
 - `type()` function shows the data type of the variable.
 - Python dynamically assigns data types.

3. Taking User Input and Printing Personalized Message

```
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# 🚀 Purpose: Taking input from user, converting types, and printing output.  
  
name = input("May I know your name please?")  
age = int(input("That's Awesome! May I get to know how old you are?"))  
fav_num = int(input("Interesting! What's your lucky number?"))  
  
print(f"Hello {name}, You are a {age} year old amazing person and your lucky number is: {fav_num}")  
print('\n') # To give a line space  
print("A Cool Note to you is - ")  
  
◀────────────────────────────────────────────────────────────────────────────────▶▶
```

- Explanation:
 - `input()` reads a string by default.
 - `int()` converts input to integer.
 - `f-string` used for clean string formatting.

4. Conditional Statements Based on Age

```
python                                                                    Copy Edit

# 🚀 Purpose: Print a message based on the user's age.

if age < 30:
    print("You are an awesome young adult who has a lot to do in life!")
elif 30 <= age <= 50:
    print("You have done a lot! Time to take rest now.")
elif 50 <= age <= 60:
    print("Transitioning from being a young person to an older person. Don't worry about transition.")
elif 60 <= age <= 80:
    print("Recreate the things which you have enjoyed.")
else:
    print("Relive every single moment, as you won't get this life back again!")
```

- **Explanation:**
 - `if-elif-else` blocks are used for multiple conditions.
 - Conditions checked sequentially until the first match.

5. Play Around with Conditions/If/Elif Loops (Machine Operating Scenario)

```
python                                                                    Copy Edit

# 🚀 Purpose: Based on age, give advice on operating a machine.

age = int(input("What's your age?"))

if age < 10:
    print("You are too young to operate this machine")
elif 11 <= age <= 30:
    print("You are at the right age to operate this machine")
elif 31 <= age <= 50:
    print("You have to upskill a lot to operate this machine")
elif 51 <= age <= 60:
    print("You should probably be just monitoring this machine by now, no need to work, just relax")
else:
    print("You are a retired person, Please take care of your amazing life ahead")
```

- **Explanation:**
 - Real-life based conditions created for better understanding.

6. Math Operations

```
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# 📌 Purpose: Perform basic mathematical operations between two numbers.

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("\n") # Just for spacing
print("Addition of two numbers is:", a + b, "\n")
print("Subtraction of two numbers is:", a - b)
print("Multiplication of two numbers is:", a * b)
print("Division of two numbers is:", a / b)
print("Mod of two numbers is:", a % b)
print("Power of two numbers is:", a ** b)
```

- **Explanation:**

- `+` addition
- `-` subtraction
- `*` multiplication
- `/` true division
- `%` modulus (remainder)
- `**` exponentiation (power)

7. Simple Calculator

```
python Copy Edit

# 📌 Purpose: Create a simple calculator that does selected operations.

num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))

print("Choose any operation which you want to operate - Addition, Subtraction, Multiplication, Division")
operation = input("")

if operation == "Addition":
    print("Result is:", num1 + num2)
elif operation == "Subtraction":
    print("Result is:", num1 - num2)
elif operation == "Multiplication":
    print("Result is:", num1 * num2)
elif operation == "Division":
    print("Result is:", num1 / num2)
else:
    print("Invalid Operation, Please try again with the options mentioned only! :)))")
```

- **Explanation:**

- Dynamic input and operation selection.
- Clean error handling for invalid operations.

Explanation Summary:

✓ Learned Concepts:

- Printing Multi-line strings using triple quotes.
- Creating and understanding variables of different types (int, float, string, boolean).
- Taking user input and type conversion (`int()` , `float()`).
- Using `f-string` for clean output.
- Writing `if-elif-else` conditional statements.
- Performing arithmetic operations.
- Building a basic calculator with user choices.

✓ New Syntax Observed:

- `"""..."""` (multi-line string)
- `type()`
- `f"{}"` (formatted string)
- `\n` (newline spacing)
- `%` (modulo)
- `**` (power)

✓ Challenges Faced:

- Confusion about `%` and `//` operators initially. (Cleared ✓)
- Clarified difference between `\n` and `""" """` usage for spacing.

✓ How I Solved It:

- Asked specific doubts on `%` , `//` , `\n` , `""" """` and now clearly understood through practice and explanation.

☀️ Self Reflection:

- Confidence Level on Today's Practice (1-10): 8.5/10
- What I Want to Improve Tomorrow:
 - Learn about loops (`for` , `while`) and functions in Python.



Mini Progress Chart

Day	Topics Covered	Confidence Level
Day 1	Printing, Variables, Input, Conditionals, Math Ops	8.5/10

✅ End of Day 1 Documentation

